CSE 107 — Midterm 2 Prep

Practice paper

Fall 2026

65 minutes | 100 points

Name: _____ **Student ID:** _____

- You may use one single-sided 8.5×11-inch page of notes and a basic nonprogrammable, nongraphing calculator. No other materials or electronic devices are permitted. Work independently.
- Answer all five questions in the spaces provided. Label any continuation with its question and part.
- For Questions 2–5, show your work. State the support of each requested PMF or density and specify that it is zero elsewhere.
- Evaluate integrals. Exact numerical expressions need not be simplified; decimals are not required.

1. (16 points)

Circle **T** or **F** for each statement. If false, give a brief reason, correction, or counterexample. True statements need no justification.

Each item is worth 2 points: a correct **T** earns 2; a correct **F** earns 1, plus 1 for a valid justification. Assume all expectations and variances mentioned below are finite and all correlations are defined.

(a) If $I_1, \dots, I_6$ are indicators with $P(I_j = 1) = 1/3$ for each j, then $\sum_{j=1}^{6} I_j \sim \text{Binomial}(6, 1/3)$. **T** **F**

(b) For any real constants a and b, $\text{Cov}(X + a, Y + b) = \text{Cov}(X, Y)$. **T** **F**

(c) If $W = 5 - 2X$, then $\text{SD}(W) = -2\,\text{SD}(X)$. **T** **F**

(d) If X is continuous, then $P(a < X \le b) = P(a \le X < b)$ for every $a < b$. **T** **F**

(e) If R and S are independent discrete random variables, then $E[R \mid S = s] = E[R]$ for every s with $P(S = s) > 0$. **T** **F**

(f) If (X, Y) has a joint density and $0 < f_Y(y) < \infty$, then $\int_{-\infty}^{\infty} f_{X|Y}(x \mid y)\, dx = f_Y(y)$. **T** **F**

(g) $\text{Cov}(X, Y) = 0$ implies $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$, even if X, Y are dependent. **T** **F**

(h) For any discrete random variables R and S, $p_{R,S}(r, s) = p_R(r)p_S(s)$ for all r, s. **T** **F**

2. (24 points)

The random variable R takes values 0 and 1, with $P(R = 1) = 1/2$. The conditional PMF of S given R is shown below; each row is zero at all other values of s.

$p_{S\|R}(s \mid r)$	$s = -1$	$s = 0$	$s = 1$
$r = 0$	1/4	1/2	1/4
$r = 1$	1/2	0	1/2

(a) Construct the complete joint PMF table of (R, S). Then find the complete marginal PMF of S. **[6 points]**

(b) Find the complete conditional PMF of R given the event $S \geq 0$. Are R and S independent? Justify your conclusion with one decisive probability comparison. **[6 points]**

(c) Find $E[R]$, $\mathrm{Var}(R)$, $E[S]$, and $\mathrm{Var}(S)$. **[6 points]**

(d) Find $E[RS]$, $\mathrm{Cov}(R, S)$, and $\mathrm{Corr}(R, S)$. Explain why your correlation is consistent with your answer about independence in (b). **[6 points]**

3. (28 points)

The pair (X, Y) has joint density

$$f_{X,Y}(x,y) = \begin{cases} 2x, & 0 < x < 1, \quad x < y < x+1, \\ 0, & \text{otherwise.} \end{cases}$$

You may use the marginal density $f_X(x) = 2x$ for $0 < x < 1$, and $f_X(x) = 0$ otherwise.

(a) Sketch the support and label its boundary lines. **[4 points]**

(b) Find the complete marginal density $f_Y(y)$. Show how the support determines the integration limits. **[8 points]**

(c) For each fixed $0 < x < 1$, find $f_{Y|X}(y \mid x)$ and $E[Y \mid X = x]$. Include the support in y of the conditional density. **[6 points]**

(d) Calculate $P(Y > 2X)$. Show the integration limits. **[6 points]**

(e) Are X and Y independent? Justify your answer. **[4 points]**

4. (16 points)

The random variable U has density

$$f_U(u) = \begin{cases} 2u, & 0 < u < 1, \\ 0, & \text{otherwise.} \end{cases} \qquad \text{Define } V = (1-U)^2.$$

(a) Use the CDF method to derive $F_V(v)$ for every real v. Express the event $\{V \le v\}$ in terms of U as part of your derivation. **[8 points]**

(b) Find a density $f_V(v)$, including where it is zero. **[2 points]**

(c) Find $E[V]$ by integration, and find $\mathrm{Var}(V)$. You may use $E[V^2] = 1/15$ without proof. **[6 points]**

5. (16 points)

The four parts below are separate problems. In $N(\mu, \sigma^2)$, the second parameter is the variance. Let Φ denote the standard normal CDF.

(a) A measurement G has distribution $N(30, 16)$. Find $P(|G - 30| \leq 6)$, showing the standardization. Leave your answer in terms of Φ. **[3 points]**

(b) A processing time T, measured in minutes, has an exponential distribution with mean 3. Find $P(T > 5 \mid T > 2)$ using the definition of conditional probability. **[4 points]**

(c) Let $U_1, \ldots, U_8$ be independent random variables, each uniform on $(-1, 3)$. Define $\overline{U} = \frac{1}{8}\sum_{i=1}^{8} U_i$. Find $E[\overline{U}]$, $\mathrm{Var}(\overline{U})$, and $\mathrm{SD}(\overline{U})$. Indicate where independence is used in the variance calculation. Standard formulas for uniform means and variances may be used. **[5 points]**

(d) Suppose $\mathrm{Var}(W) = 1$, $\mathrm{Var}(Z) = 4$, and $\mathrm{Corr}(W, Z) = -1/4$. Find $\mathrm{Var}(3W + 2Z - 7)$. **[4 points]**